

Non-Simply Connected Subsystems and the Reduced Density Matrices

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Abstract

The reduced density matrix of a multiply connected subsystem exhibits an exact diagonal structure different from that of a simply connected subsystem.

First, we consider a non-simply connected region A with an interior hole. The reduced density matrix of this subsystem is as follows,

$$\rho_A = \frac{1}{2^{N^A - N_p^A}} \hat{\Pi}_A (\mathbb{1} + W_{\mathcal{L}}) \sum_{[\mathfrak{D}] \subset A} (\mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}} + \mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{H}} \Sigma_{\mathfrak{D} \ominus \mathfrak{H}})$$

where $\mathcal{L}$ is a closed loop inside the subsystem A that encloses the hole. In other words, the loop $\mathcal{L}$ is a non-contractible loop according to the non-simply connected subsystem A . The edge configuration $\mathfrak{H}$ represents the *web-patch* configuration of the hole enclosed by A consisting of all the dimers inside the hole and those that cross the inner boundary.

$$\begin{aligned} \rho_A &= \frac{1}{2^{N^A - N_p^A - 1}} \hat{\Pi}_A \left(\frac{\mathbb{1} + W_{\mathcal{L}}}{2} \right) \sum_{[\mathfrak{D}] \subset A} \left(\mathbb{1} + \underbrace{\varphi(\mathfrak{H}, \mathfrak{D}) \frac{\mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{H}}}{\mathfrak{C}_{\mathfrak{D}}}}_{\equiv 2p_{\mathfrak{D}} - 1} \Sigma_{\mathfrak{H}} \right) \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}} \\ \rho_A &= \frac{1}{2^{N^A - N_p^A - 2}} \hat{\Pi}_A \hat{\Pi}_{\mathcal{L}} \sum_{[\mathfrak{D}] \subset A} \left[p_{\mathfrak{D}} \left(\frac{\mathbb{1} + \Sigma_{\mathfrak{H}}}{2} \right) + (1 - p_{\mathfrak{D}}) \left(\frac{\mathbb{1} - \Sigma_{\mathfrak{H}}}{2} \right) \right] \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}} \\ \rho_A &\equiv \frac{1}{2^{N^A - N_p^A - 2}} \hat{\Pi}_A \hat{\Pi}_{\mathcal{L}} \left[\left(\hat{\Pi}_{\mathfrak{H}}^{(+)} \sum_{[\mathfrak{D}] \subset A} \tilde{\mathfrak{C}}_{\mathfrak{D}}^{(+)} \Sigma_{\mathfrak{D}} \right) \oplus \left(\hat{\Pi}_{\mathfrak{H}}^{(-)} \sum_{[\mathfrak{D}] \subset A} \tilde{\mathfrak{C}}_{\mathfrak{D}}^{(-)} \Sigma_{\mathfrak{D}} \right) \right] \\ p_{\mathfrak{D}} &= \frac{1}{2} \left(1 + \varphi(\mathfrak{H}, \mathfrak{D}) \frac{\mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{H}}}{\mathfrak{C}_{\mathfrak{D}}} \right) \\ \tilde{\mathfrak{C}}_{\mathfrak{D}}^{(+)} + \tilde{\mathfrak{C}}_{\mathfrak{D}}^{(-)} &= \mathfrak{C}_{\mathfrak{D}} \\ \begin{cases} \tilde{\mathfrak{C}}_{\emptyset}^{(+)} = p_{\emptyset} = \frac{1 + \mathfrak{C}_{\mathfrak{H}}}{2} \\ \tilde{\mathfrak{C}}_{\emptyset}^{(-)} = 1 - p_{\emptyset} = \frac{1 - \mathfrak{C}_{\mathfrak{H}}}{2} \end{cases} \\ \hat{\Pi}_{\mathfrak{H}}^{(\pm)} &= \frac{\mathbb{1} \pm \Sigma_{\mathfrak{H}}}{2} \end{aligned}$$

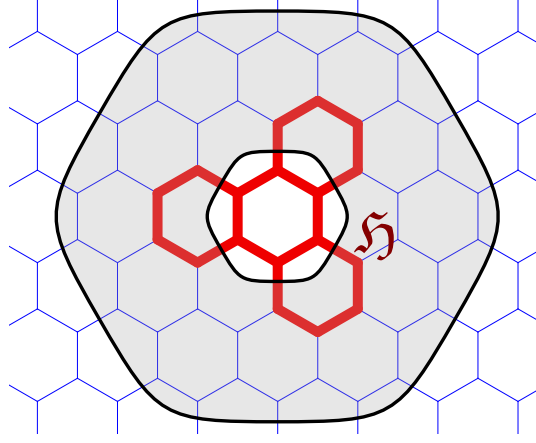


Figure 1: The web-patch configuration $\mathfrak{H}$ covering the interior hole of a non-simply connected subsystem. Its corresponding string operator $\Sigma_{\mathfrak{H}}$ commutes with other string operators lying within the shaded subsystem.

$$\text{Tr}_A \left(\hat{\Pi}_A \hat{\Pi}_{\mathcal{L}} \hat{\Pi}_{\mathfrak{H}}^{(\pm)} \right) = 2^{N^A - N_p^A - 2}$$

$$\left(\text{Diagram 1} \right) = \left(\text{Diagram 2} \right) \ominus \left(\text{Diagram 3} \right)$$

Diagram 1: A large gray region with a black-outlined hole. Inside the hole is a red hexagonal patch labeled $\mathfrak{H}$.
Diagram 2: A large gray region with a black-outlined hole. Inside the hole is a red hexagonal patch labeled $\mathfrak{H}$.
Diagram 3: A large gray region with a black-outlined hole. Inside the hole is a red hexagonal patch labeled $\mathfrak{H}$.

$$\rho_A = \frac{1}{2^{N^A - N_p^A - 2g}} \hat{\Pi}_A \hat{\Pi}_g \bigoplus_{\{s\}} \left[\hat{\Pi}_{\mathfrak{H}}^{(\{s\})} \sum_{[\mathfrak{D}] \subset A} \tilde{\mathfrak{C}}_{\mathfrak{D}}^{(\{s\}) \Sigma_{\mathfrak{D}}} \right] \quad (0.1)$$

$$\begin{cases} \hat{\Pi}_g \equiv \prod_{n=1}^g \left(\frac{\mathbb{1} + W_{\mathcal{L}^{(n)}}}{2} \right) \\ \hat{\Pi}_{\mathfrak{H}}^{(\{s\})} \equiv \prod_{n=1}^g \left(\frac{\mathbb{1} + s_n \Sigma_{\mathfrak{H}^{(n)}}}{2} \right) \end{cases} \quad (0.2)$$

$$\sum_{\{s\}} \tilde{\mathfrak{C}}_{\mathfrak{D}}^{(\{s\})} = \mathfrak{C}_{\mathfrak{D}} \quad (0.3)$$